

Twenty-five conjectures from a local–global random model: the statements

1 Notation

p, q denote primes; γ is Euler’s constant, $e^\gamma = 1.781072\dots$; φ is Euler’s totient; $\phi = (1 + \sqrt{5})/2$; F_n the Fibonacci numbers; $p\#$ the primorial. For irreducible $f_1, \dots, f_k \in \mathbb{Z}[x]$ with positive leading coefficients, the Bateman–Horn singular series is

$$C(f_1, \dots, f_k) = \prod_p \frac{1 - \omega(p)/p}{(1 - 1/p)^k}, \quad \omega(p) = \#\{n \bmod p : p \mid f_1(n) \cdots f_k(n)\},$$

and the corresponding main term is

$$I(N) = \int_2^N \frac{dt}{\prod_i \log f_i(t)},$$

which absorbs the usual $1/\prod_i \deg f_i$. The Bateman–Horn conjecture [2] asserts $\#\{n \leq N : \text{all } f_i(n) \text{ prime}\} \sim C \cdot I(N)$ whenever $C \neq 0$; each Bateman–Horn conjecture below is an instance with its constant evaluated. For nonlinear systems the product converges only conditionally; throughout, it is taken in the canonical ordering of increasing primes, in which convergence follows from the equidistribution of $\omega(p)$ about its Chebotarev mean. Our numerical truncation wobble is an error bar for this canonically ordered product. We write $\text{Li}_k(x) = \int_2^x (\log t)^{-k} dt$. The twin singular series is $\mathfrak{S}(d) = 2C_2 \prod_{p|d, p>2} \frac{p-1}{p-2}$ for even d , with $2C_2 = 1.3203236\dots$; the Goldbach series $\mathfrak{S}(n)$ is the same product over odd $p \mid n$.

Eight of the conjectures below are built over the Bateman–Horn move—choose an admissible system, compute its singular series, state the count—but none is a bare instance of it: 10 and 15 are family laws with derived constant distributions, 23 and 25 carry structural classifications, 24 is a searched chain, and 18, 19, 14 are contamination-calculus predictions (two fresh races and the negative control) whose mechanism is the contamination calculus of Conjecture 1(v). Each carries its classical calibration instance as an attributed remark. The heuristic is uniform, so we give it once. Model the primality of the values $f_i(n)$ as independent events of probability $1/\log f_i(n)$, corrected by the factor $(1 - \omega(p)/p)/(1 - 1/p)^k$ at each prime p to account for the joint local behaviour; multiplying the corrections and integrating gives $C \cdot I(N)$.

For locally integrable F we write $\mathcal{M}_x(F) = \frac{1}{\log x} \int_2^x F(t) \frac{dt}{t}$ for the logarithmic mean; all race conjectures below state their drift and null clauses through $\mathcal{M}_x$, so that “no persistent leader” and “systematic surplus” are claims about a defined functional rather than about pointwise behaviour. Drift clauses are stated at the *drift scale*—for a pair race, $\mathcal{M}_x(D(t) \log^2 t / \sqrt{t})$ converging to an explicit constant—never as $\mathcal{M}_x((D - T)/\sqrt{\pi}) \rightarrow 0$, which is vacuous for the drift coefficient (both $T/\sqrt{\pi}$ and its multiples have logarithmic mean $\rightarrow 0$). Convergence of $\mathcal{M}_x$ at drift scale is itself part of each such conjecture: a pure random-walk fluctuation model would make these means

divergent in variance, so their convergence encodes the Rubinstein–Sarnak-type almost-periodicity that the zero-oscillation hypothesis names.

Admissibility ($\omega(p) < p$ for all p) is checked for every constant computed. The natural-looking pair $\{n^2 + n + 1, n^2 + n + 3\}$ has $\omega(3) = 3$ and is rejected; the $k = 4$ branch of Conjecture 25 and the naive iterated chain behind Conjecture 24 are inadmissible for the same reason, and both exclusions are part of the statements.

2 Principal conjectures

Lemma 1 (Orientation elimination and class assignment). *In $\psi_2(x) = \sum_{n \leq x} \Lambda(n)\Lambda(n+2)$, the total contribution of terms in which n or $n+2$ is a proper prime power is, apart from $O(x^{1/3} \log^2 x)$ from cubes and higher powers, carried entirely by the patterns $(q^2 - 2, q^2)$ with q prime and $q^2 - 2$ prime: the mirror pattern $(q^2, q^2 + 2)$ is annihilated by $3 \mid q^2 + 2$ for every prime $q > 3$. Moreover, for every prime $q \neq 5$, $q^2 - 2 \equiv 2$ or $4 \pmod{5}$ (never 1), and for every odd q , $q^2 - 2 \equiv 7 \pmod{8}$. (The single exceptional term $q = 5$, $(23, 25)$, lands outside the twin-start classes altogether and is absorbed in the bounded error.)*

Conjecture 1 (Twin races modulo 5 and 8, and the prime-power contamination calculus; apparently new). *Twin starts $p > 5$ lie in classes $p \equiv 1, 2, 4 \pmod{5}$. Let $D_1(x) = \pi_t(x; 5, 1) - \frac{1}{2}(\pi_t(x; 5, 2) + \pi_t(x; 5, 4))$ and*

$$T(x) = \frac{1}{2 \log^2 x} \sum_{\substack{q \leq \sqrt{x} \\ q^2 - 2 \text{ prime}}} \log q \log(q^2 - 2),$$

whose scale is governed by the constant C_7 of Conjecture 19. Then: (i) [drift law, at drift-scale normalization] the clause asserts two sub-claims, labelled separately because they can fail independently. (i-a) [mechanism] class 1 carries the systematic surplus T : one has the decomposition $D_1(t) = T(t) + R(t)$ in which the remainder R carries no deterministic component at the drift scale, so that the entire drift-scale deterministic content of D_1 is the contamination term and its coefficient is thereby identified. (i-b) [averaging] at the drift-scale normalization the remainder averages away:

$$\mathcal{M}_x \left(\frac{R(t) \log^2 t}{\sqrt{t}} \right) \longrightarrow 0, \quad \text{equivalently} \quad \mathcal{M}_x \left(\frac{D_1(t) \log^2 t}{\sqrt{t}} \right) \longrightarrow c_T,$$

where

$$c_T := \lim_{x \rightarrow \infty} \frac{T(x) \log^2 x}{\sqrt{x}} = \frac{1}{2} C(x, x^2 - 2)$$

is the half of the Bateman–Horn constant of the pair $(q, q^2 - 2)$ (the weighted census sum is $\sim C\sqrt{x}$, so the limit exists and is explicit). A normalization at the noise scale, $\mathcal{M}_x((D_1 - T)/\sqrt{\pi_t}) \rightarrow 0$, would be vacuous for the coefficient of (i-a): since $T/\sqrt{\pi_t} \asymp 1/\log t$, one has $\mathcal{M}_x(T/\sqrt{\pi_t}) \asymp \log \log x / \log x \rightarrow 0$, so the condition holds with T replaced by 0, $2T$, or any fixed multiple, and does not express “exactly the removed mass.” The statement therefore normalizes at the drift scale, where the claimed limit is a nonzero constant. Convergence at this normalization is a Rubinstein–Sarnak-type regularity assertion: under a pure random-walk model the logarithmic mean would not converge (its variance grows like $\log x$), so (i-b) asserts almost-periodic structure for the twin

race and is the precise content of the zero-oscillation hypothesis, while (i-a) is what fixes the limit at c_T rather than 0 or $2c_T$, and is the calculus. The convergence asserted in (i-b) selects the rigid branch of the dichotomy registered at Conjecture 9(iii); the two statements stand or fall together at drift scale; (ii) [zero deterministic drift in the symmetric differences] the symmetric classes carry no deterministic term at the same normalization: $\mathcal{M}_x((\pi_t(\cdot; 5, 2) - \pi_t(\cdot; 5, 4)) \log^2 t / \sqrt{t}) \rightarrow 0$, and likewise for each of the three pairwise differences among classes $\{1, 3, 5\} \bmod 8$ in clause (iv)—now a statement distinct from (i), since at drift scale “zero” and “ c_T ” are different limits; (iii) [sign density, conditional] conditional on the Gaussian fluctuation model for the normalized remainder in (i)–(ii), the logarithmic density of $\{D_1 > 0\}$ exists and equals $\frac{1}{2}$, approached from above with a finite- x excess of order $\log \log x / \log x$: the pointwise drift-to-noise ratio is $T / \sqrt{\pi_t} \asymp 1 / \log t$, and its logarithmic average $\mathcal{M}_x(1 / \log t) \sim \log \log x / \log x$ carries an extra $\log \log$ (the scale is not $1 / \log x$). In contrast to Chebyshev’s races, the twin race has no persistent leader; (iv) [mod-8 companion] since $q^2 - 2 \equiv 7 \pmod{8}$ always (Lemma 1), the entire deterministic term falls on the single class $7 \pmod{8}$: with $D_7(x) = \frac{1}{3} \sum_{a \in \{1, 3, 5\}} \pi_t(x; 8, a) - \pi_t(x; 8, 7)$,

$$\mathcal{M}_x\left(\frac{D_7(t) \log^2 t}{\sqrt{t}}\right) \rightarrow 2c_T$$

(twice the mod-5 constant, undiluted—at this normalization the factor 2 is a falsifiable prediction, the mechanism and averaging claims separating here exactly as in (i-a)–(i-b)), while classes 1, 3, 5 are mutually symmetric. The mod-5 and mod-8 races are two projections of one mechanism, so their joint behaviour is a family test in the sense of the shape-of-agreement criterion; (v) [contamination calculus for balanced races] for a pattern $(n, n + d)$ and modulus m , call the race balanced if the prime-prime singular series is identical across the competing residue classes (as CRT gives for the classes compared in (i)–(iv)) and no mechanism other than prime powers contributes at the $\sqrt{x}$ scale under the zero-oscillation hypothesis. For balanced races the deterministic drift vector is computed by one operator. Define the contamination operator

$$\mathcal{C}_{d,m}(a; x) = \sum_{\text{orientations } o \in \{(q^2-d, q^2), (q^2, q^2+d)\}} \sum_{\substack{q \leq \sqrt{x}, \text{ } o \text{ prime-compatible} \\ \text{start}(o) \equiv a \pmod{m}}} \Lambda(q^2) \Lambda(\text{prime member of } o),$$

where prime-compatible means the non-square member is prime and no fixed prime annihilates the orientation. The provable layer is the orientation census and class assignment (which o survive, and where their starts land—Lemma 1 and its analogues); the conjectural layer is the transfer: the unweighted class counts are depressed by $\mathcal{C}_{d,m}(a; x) / \log^2 x$ relative to the clean classes, in the drift-scale limiting-log-mean sense of (i), with the mechanism and averaging claims separated as in (i-a)–(i-b)—for each class a , the deficit obeys $\mathcal{M}_x(\text{deficit}_a(t) \log^2 t / \sqrt{t}) \rightarrow c(a) := \lim \mathcal{C}_{d,m}(a; x) / \sqrt{x}$, an explicit Bateman–Horn constant per class, so every coefficient in the vector is identified—i.e. “exactly the removed mass,” which requires the weighted identity, partial summation uniform in classes, higher-power bounds (exponents $r \geq 3$ contribute $O(x^{1/3+\epsilon})$ against the operator’s $\asymp \sqrt{x}$ and are negligible—the one easy layer of the transfer), and the zero-oscillation hypothesis, and is the calculus’s single analytic conjecture rather than a rule. The drift is $\asymp \sqrt{x} / \log^2 x$ against a count fluctuation $\asymp \sqrt{x} / \log x$, i.e. smaller by $1 / \log x$ in standard-deviation units at every height—so the calculus predicts occupation biases under long logarithmic aggregation, never a fixed nonzero standardized mean, and every verification in this family is designed around that fact. (Races with unequal primary constants, or with interfering orientations beyond those enumerated, are outside the calculus until those terms are separated.) In particular, for the

cousin pattern $d = 4$ the calculus makes two predictions with no free choices: the orientation $(q^2 - 4, q^2)$ is dead ($q^2 - 4 = (q - 2)(q + 2)$ is composite), the orientation $(q^2, q^2 + 4)$ survives (no prime annihilates it), $q^2 + 4$ prime forces $q \equiv \pm 2 \pmod{5}$ for $q \neq 5$ (at $q = 5$ the value 29 is prime, but the start $25 \equiv 0 \pmod{5}$ lies outside the cousin start classes $\{2, 3, 4\}$), so the contaminated cousin-start class is $4 \pmod{5}$ (of $\{2, 3, 4\}$) and—since $q^2 \equiv 1 \pmod{8}$ —the class $1 \pmod{8}$ (of $\{1, 3, 5, 7\}$): both deficits converge, in the drift-scale limiting-log-mean sense of (i)–(ii), to $c^c = \lim_{x \rightarrow \infty} T^c(x) \log^2 x / \sqrt{x}$, the Bateman–Horn constant of the pair $(q, q^2 + 4)$, where $T^c(x) = \frac{1}{\log^2 x} \sum_{q \leq \sqrt{x}, q \neq 5, q^2 + 4 \text{ prime}} \log q \log(q^2 + 4)$.

Conjecture 2 (The pair-level Montgomery–Soundararajan reduction; apparently new). *For the window field of Conjecture 8(ii), the diagonal variance of the twin-pair count reduces to a pinned singular-series average. With $R(h) = C_4(0, 2, h, h + 2)/\mathfrak{S}(2)^2$ for $h \geq 1$ (zero when the offset set is degenerate or inadmissible—all odd h , and $h = 2$ since $\{0, 2, 4\}$ dies at 3; consecutive twin pairs cannot overlap beyond $(3, 5, 7)$, so no overlap term exists) and*

$$G(H) = \sum_{1 \leq h \leq H} \left(1 - \frac{h}{H}\right) (R(h) - 1),$$

the derived identity is

$$\frac{\text{Var}_t \pi_2(t; H)}{\mathbb{E}_t \pi_2(t; H)} = 1 + \frac{\mathfrak{S}(2) (2G(H) - 1)}{\log^2 x} + o\left(\frac{1}{\log^2 x}\right) :$$

with $p = \mathfrak{S}(2)/\log^2 x$ the per-site intensity, the variance is $Hp(1 - p) + 2p^2HG(H)$, so $\text{Var}/\mathbb{E} = 1 - p + 2pG(H)$ —the Bernoulli diagonal term $-p = -\mathfrak{S}(2)/\log^2 x$ is of exactly the order of the claimed error and cannot be dropped (it is the same diagonal that, in the single-prime case, merges into Montgomery–Soundararajan’s constant $\gamma + \log 2\pi - 1$). With that term retained, the entire deficit left open at Conjecture 8(ii) is the single computable function G plus the explicit Bernoulli correction—the pair analogue of Montgomery–Soundararajan’s reduction of the prime-count variance to $\sum_{h \leq H} \mathfrak{S}(h)(H - h)$ [19]. Clause (i) is a conditional proposition, not an independent conjecture: given quantitative Hardy–Littlewood for 4-tuples, the reduction follows. The conjectural clause is (ii): $-G(H)/\log H$ diverges—the pinned average grows strictly faster than the single-prime secondary term $\frac{1}{2}(\log H + \gamma + \log 2\pi - 1)$, so pair counts in windows are more sub-Poisson than prime counts—with the caveat that data to $H = 3000$ cannot distinguish divergence from a pure log with a large constant (nor $c \log H \log \log H$ from $(\log H)^\alpha$): what the computation establishes is only that the local slope exceeds the single-prime value sevenfold across the computable range; the divergence and its exact form are registered, not evidenced beyond that.

Conjecture 3 (Least primes in progressions; part (i) previously stated [20, 21, 30, 29], part (ii) apparently new). *Let $p(a, q)$ be the least prime $\equiv a \pmod{q}$ and $U(a, q) = \text{Li}(p(a, q))/\varphi(q)$. Then: (i) $U \Rightarrow \text{Exp}(1)$ marginally as $q \rightarrow \infty$; and, under the additional hypothesis that the class processes are asymptotically jointly independent in the extreme-value sense (which marginal convergence alone does not supply), $\max_a U - H_{\varphi(q)}$ converges to a Gumbel law ($H_n = \sum_{k \leq n} 1/k$), the order-statistics form of Wagstaff’s $\max_a p(a, q) \sim \varphi(q) \log^2 q$; (ii) [canonical deficit, dyadic-average form] define, with no reference to any control model, $\Theta(q) = (1 - \mathbb{E}_a[U(a, q)]) \log q$. The conjecture is stated for dyadic averages over prime moduli—pointwise convergence along every prime q may be too strong, since arithmetic fluctuations in q could persist:*

$$\frac{1}{\#\{q \in [Q, 2Q] \text{ prime}\}} \sum_{\substack{Q \leq q \leq 2Q \\ q \text{ prime}}} \Theta(q) \rightarrow \Theta > 0 \quad (Q \rightarrow \infty).$$

(The formal claim is model-free: existence and strict positivity of the dyadic-average limit. The stronger reading—that Θ exceeds the discreteness level any parity-aware control produces—is the diagnostic layer below, not part of the display, since a control constant has no place in a model-free statement.) The limiting distribution of $\Theta(q)$ over the dyadic block is left open, and in particular we do not assert $\text{sd}(\Theta(q)) \rightarrow 0$: fluctuations driven by the arithmetic of q and of $q - 1$ (through the singular-series pair terms $\mathfrak{S}(kq)$ and the class structure mod q) may well persist at bounded size, so $\Theta(q)$ need not concentrate around its dyadic mean.

Conjecture 4 (The Goldbach lane race; apparently new). For $n \equiv 2 \pmod{4}$ let $R_1(n)$, $R_3(n)$ be the ordered Goldbach representation counts in the lanes $p \equiv q \equiv 1$ and $p \equiv q \equiv 3 \pmod{4}$ (Conjecture 5), and $D(n) = R_3(n) - R_1(n)$. Prime-square terms $n = q^2 + p$ (q odd prime) land exclusively in the $(1, 1)$ lane, since $q^2 \equiv 1 \pmod{4}$ and then $n - q^2 \equiv 1 \pmod{4}$ automatically. Consequently the $(3, 3)$ lane leads on average: with

$$D_{\text{sys}}(n) = \frac{2}{\bar{\ell}(n)} \sum_{\substack{q \text{ odd prime, } q \leq \sqrt{n} \\ n - q^2 \text{ prime}}} \log q \log(n - q^2), \quad \bar{\ell}(n) = \frac{n - 6}{\int_3^{n-3} \frac{dt}{\log t \log(n - t)}},$$

(the analytic mean of $\log p \log(n - p)$ under the lane profile; the empirical lane mean is its estimator). Write $\mathbb{E}_x$ for the average over n drawn from $\{n \equiv 2 \pmod{4}, n \leq x\}$ with weight proportional to $1/n$ (the logarithmic sampling measure, now explicit). Then: (i) [drift law] $\mathbb{E}_x[D - D_{\text{sys}}] = o(\mathbb{E}_x[D_{\text{sys}}])$ as $x \rightarrow \infty$: the $(3, 3)$ lane leads on average by exactly the square contamination; (ii) [weighted mean and internal null lane] $\mathbb{E}_x[D_{\text{sys}}]$ is given by the Hardy–Littlewood (Conjecture H) local model for $n - q^2$, whose local factors vary strongly with n —most drastically at $p = 3$: for $n \equiv 1 \pmod{3}$, $3 \mid n - q^2$ for every prime $q > 3$, so the contamination collapses to exactly two boundary families—the $q = 3$ term (when $n - 9$ is prime) and—a second family, not a single term—the sparse exceptional family $n - q^2 = 3$ (when $n - 3$ is itself a prime square, so that the divisible-by-3 value is 3)—both negligible in logarithmic mean, so the race carries no drift at the systematic scale on the subprogression $n \equiv 10 \pmod{12}$: with the positive comparison scale $A_G(x) := \mathbb{E}_x^{(\neq 1 \pmod{3})}[D_{\text{sys}}]$, the logarithmic-ensemble mean of the systematic term over the contaminated lane $n \not\equiv 1 \pmod{3}$ (a comparison against this subprogression’s own systematic term, which vanishes, would be empty), the claim is $\mathbb{E}_x[D] = o(A_G(x))$ on $n \equiv 10 \pmod{12}$, while the drift concentrates on $n \not\equiv 1 \pmod{3}$: an internal null lane, predicted by the same mechanism that predicts the lead; (iii) [sign density] the per- n drift-to-noise ratio $\kappa(n) = D_{\text{sys}}(n)/\sqrt{\mathfrak{S}(n)J(n)}$ (where $J(n) = \int_3^{n-3} dt/(\log t \log(n - t))$, so that $\mathfrak{S}(n)J(n)$ is the total ordered representation count) satisfies $\kappa(n) \asymp c(n)/\log n \rightarrow 0$; under the Gaussian fluctuation model the logarithmic density of $\{D > 0\}$ is therefore $\frac{1}{2}$ in the limit, with the finite- x sign fraction predicted to be $\mathbb{E}_x[\Phi(\kappa)]$ (Φ the standard normal distribution function)—decaying, but far from $\frac{1}{2}$ at any accessible height.

Conjecture 5 (The least Goldbach summand: exponential law and ordering deficit; the time-changed law and deficit constant apparently unstated). Let $s(n)$ be the least prime $p \nmid n$ with $n - p$ prime (the exclusion $p \nmid n$ removes a trivial obstruction: $p \mid n$ forces $p \mid n - p$, so $n - p$ is composite except in the diagonal case $n = 2p$, which would otherwise distort the hazard at one atypical point), and time-change by the expected-arrivals clock $U(n) = \mathfrak{S}(n) \sum_{p \leq s(n), p \nmid n} 1/\log(n - p)$. Then: The ensemble is fixed first, since $U(n)$ is deterministic at each n and an expectation therefore needs a declared measure: for a scale X ,

$$\mathbb{E}_X U = \frac{\sum_{X < n \leq 2X, 2 \mid n} U(n)/n}{\sum_{X < n \leq 2X, 2 \mid n} 1/n}, \quad \Theta_G(X) = (1 - \mathbb{E}_X U) \log X.$$

(i) [limit law] under $\mathbb{E}_X$ -sampling, $U \Rightarrow \text{Exp}(1)$ as $X \rightarrow \infty$; (ii) [canonical ordering deficit] $\Theta_G(X) \rightarrow \Theta_G > 0$ —the Goldbach sibling of the least-prime deficit of Conjecture 3(ii), produced by the same occupancy mechanism: the ordered primes fill the available representations faster than an exchangeable model, so the first arrival comes early and $\mathbb{E}_X[U] < 1$.

Conjecture 6 ($n^2 + 2^n$; sequence is OEIS A064539). Every prime of the form $n^2 + 2^n$ with $n > 1$ has $n \equiv 3 \pmod{6}$ (elementary), and: (i) infinitely many such primes exist, with counting function $\asymp \log N$; (ii) [candidate law, entanglement-aware form] for a finite set S of primes $p \geq 5$ (the primes 2 and 3 are exact in the lane reduction, $\delta_2 = \delta_3 = 0$, and $\text{ord}_2 2$ is undefined) let D_S be the exact joint density, within the lane $n \equiv 3 \pmod{6}$, of n with $p \nmid n^2 + 2^n$ for all $p \in S$, computed over the full joint period $\text{lcm}_{p \in S} \text{lcm}(6, p \cdot \text{ord}_p 2)$, and set $\kappa_S = D_S / \prod_{p \in S} (1 - 1/p)$. The conjecture is stated along the canonical exhaustion $S_z = \{p \text{ prime} : 5 \leq p \leq z\}$, and its three layers are logically separate and are asserted separately: (ii-a) the sequence κ_{S_z} converges as $z \rightarrow \infty$ (a general net over arbitrary finite S carries an order-of-exhaustion ambiguity we do not need), with limit κ ; (ii-b) whether the limit factors—i.e. whether the exact finite-level factorization observed below persists, so that entanglement is asymptotically absent—is an open question, expressly not counted among this paper’s conjectural assertions, on which (ii-a) takes no position; and (ii-c) the count is $\sim \sum_{n \leq N, n \equiv 3 \pmod{6}} \kappa / \log(n^2 + 2^n)$ —a genuine further step, since a convergent local density does not automatically govern the global count. The working value $\kappa^* = 4.2734\dots$ is a hybrid, not a computation of any single κ_{S_z} : the CRT-exact core S_{19} times independent per-prime factors for $19 < p \leq 300$ —so $\kappa^* = \kappa_{S_{300}}$ exactly if and only if the observed exact factorization persists to 300.

Conjecture 7 (Fermat quotients: the multibase subtorus law and the Wieferich ledger; single-base heuristic previously stated [22], the vertical multibase law apparently new). With $q_p(a) = (a^{p-1} - 1)/p \pmod{p}$ (defined for primes $p \nmid a$; all clauses below range over such p only) and $q_p = q_p(2)$, five clauses: (0) [structure, classical] $q_p(ab) \equiv q_p(a) + q_p(b) \pmod{p}$ (the Eisenstein–Lerch homomorphism; verified exactly on every tested prime), so for multiplicatively dependent bases the vector $(q_p(a_1), \dots, q_p(a_r))/p$ is confined to the rational subtorus cut out by the relations; (iv) [multibase equidistribution, the claimed law] for multiplicatively independent bases the vector equidistributes on the full torus as p varies—the vertical joint law, complementing the fixed- p , varying-base statistics of Ostafe–Shparlinski and Cobeli–Zaharescu—with correlations vanishing and the same LIL calibration as (i); (v) [simultaneous Wieferich] the expected count of $p \leq x$ with $q_p(2) = q_p(3) = 0$ has convergent sum $\sum 1/p^2$: at most finitely many simultaneous Wieferich primes exist (the single-base folklore of this accounting is Conrad’s), and the empirical list is empty to 10^7 . Three further clauses on the base-2 quotient alone: (i) [global equidistribution] q_p/p equidistributes on $[0, 1)$ with discrepancy at exactly the calibrated random-model scale: in its sharp form,

$$\limsup_{x \rightarrow \infty} \frac{\sqrt{\pi(x)} D_{KS}(x)}{\sqrt{\log \log \pi(x)}} = \frac{1}{\sqrt{2}},$$

the Chung–Smirnov law-of-the-iterated-logarithm constant for an i.i.d. uniform sequence—demanding both no more (a bound $O(\pi(x)^{-1/2})$ would be stronger than the random model supports, the same over-demand that killed the Mertens conjecture) and no less than the model earns: the upper bound alone would be consistent with hidden rigidity, and the matching lower bound asserts the quotients fluctuate like genuine noise; (ii) [shrinking targets] for every fixed $K \geq 1$, and uniformly for $K = K(x) \leq \log x$, $\#\{p \leq x : q_p < K\} \sim \sum_{p \leq x} \min(1, K/p)$ —the fixed- K case down to $K = 1$ is what governs the zero event and does NOT follow from (i), since $\{q_p = 0\}$ is a target

of shrinking measure $1/p$; (iii) [Wieferich count, fluctuation-calibrated, with the probability space declared] $W(x) = \#\{\text{Wieferich } p \leq x\}$ is a deterministic function, so its fluctuation clauses are stated in the two legitimate forms, an unqualified CLT for a fixed sequence having no meaning: the deterministic envelope conjecture

$$\limsup_{x \rightarrow \infty} \frac{W(x) - \log \log x}{\sqrt{2 \log \log x \log \log \log x}} = 1$$

(the iterated-logarithm scale at variance $V(x) \sim \log \log x$; a bound $O_\varepsilon(V^{1/2+\varepsilon})$ would be weaker than the model's exact prediction), and the almost-sure central limit law, at the correct logarithmic weighting: with $u = \log \log x$ the variance-time of the count, sample u with density du/u on $[e, L]$, $L = \log \log X$ (equivalently: $\log u = \log \log \log x$ uniform); then the sampled distribution of $(W(x) - u)/\sqrt{u}$ converges to $N(0, 1)$ as $X \rightarrow \infty$. The weighting is what makes the clause true, and the two natural alternatives both fail. A block-sampled version on $[X, X^A]$ fails because only $\log A = O(1)$ new events enter the block. A uniform-in- u version on $[1, L]$ fails too, by Brownian scaling: with $W(e^u) - u \approx B(u)$ and $u = Ls$, the self-similarity $B(Ls)/\sqrt{Ls} \stackrel{d}{=} \tilde{B}(s)/\sqrt{s}$ means the uniform- u empirical distribution converges not to Φ but to the random occupation functional $\int_0^1 \mathbf{1}\{\tilde{B}(s)/\sqrt{s} \leq z\} ds$ —values at proportional times stay correlated, so uniform sampling never averages over enough independent scales. The du/u weighting is the classical almost-sure-CLT weighting (the continuous analogue of the $1/k$ weights), for which the model does predict almost-sure convergence of the sampled law to Φ . Both clauses are conjectured on the Crandall–Dilcher–Pomerance side of the model choice that clause (iv)'s discussion records.

Conjecture 8 (Uniform quantitative de Polignac; the core is previously stated). (i) $\pi_d(x) = \#\{p \leq x - d : p, p + d \text{ prime}\} = \mathfrak{S}(d) \text{Li}_2(x) (1 + o(1))$ uniformly for even $d \leq (\log x)^A$. (ii) [residual field, with derived covariance kernel] The same primes participate in many pair counts, and the covariance of two of them is carried by triple configurations sharing one prime. For even $d \neq d'$ define the overlap constant

$$K(d, d') = C_3(0, d, d') + C_3(0, d', d + d') + C_3(0, d, d + d') + C_3(0, |d - d'|, \max(d, d')),$$

where C_3 denotes the Hardy–Littlewood triple constant of the offset set (degenerate or inadmissible triples contributing 0). The field is randomized canonically—covariances of the deterministic cumulative counts are otherwise undefined: for t uniform on $[x, 2x]$ and a window length H , let $\pi_d(t; H) = \#\{t < n \leq t + H : n, n + d \text{ prime}\}$ and let $Z_d(t; H)$ be its studentization. The mean count per window is $\asymp H/\log^2 x$, and the regime is set by that ratio (a phase condition: a bare $H \rightarrow \infty$ does not fix the regime):

$$\text{Cov}_t(\pi_d, \pi_{d'}) \sim \frac{K(d, d') H}{\log^3 x}, \quad \rho(d, d') \sim \frac{K(d, d')}{\sqrt{\mathfrak{S}(d)\mathfrak{S}(d')} \log x},$$

valid for $\max(d, d') = o(H)$, which is the standing restriction of this clause and holds throughout the window class fixed below. In general each overlap orientation o —the in-window displacements $h_o \in \{0, d, -d', d - d'\}$ and their reflections—enters with multiplicity $(H - |h_o|)_+$ rather than H , so that the shared-prime term reads

$$\frac{1}{\log^3 x} \sum_o (H - |h_o|)_+ K_3^{(o)}(d, d'), \quad \sum_o K_3^{(o)}(d, d') = K(d, d'),$$

and an orientation with $|h_o| > H$ is absent from the window altogether; under $\max(d, d') = o(H)$ one has $(H - |h_o|)_+ = H(1 + o(1))$ for every orientation and the displayed form is recovered, so the uniform-in- d statement carries that restriction throughout. Here the shared-prime (triple) term is the first covariance contribution but not the whole of it: the full expansion is

$$\text{Cov}_t(\pi_d, \pi_{d'}) = \frac{K(d, d') H}{\log^3 x} + \frac{1}{\log^4 x} \sum_{|h| \leq H} (H - |h|) K_4(d, d'; h) + \text{error},$$

with K_4 the centred singular series of the off-index four-point sets $\{0, d, h, h + d'\}$ —the cross analogue of the pinned diagonal average of Conjecture 2—where degenerate offset relations are handled inside the kernels: whenever the four-point set collapses (repeated entries, as at $h = 0$, $h = d$, $h = -d'$, or $d' - d = \pm h$) the configuration is a triple or pair and its contribution belongs to the K -term (or to the mean), so K_4 is defined as the centred singular series of genuinely four-element sets, with special pairs such as $d' = 2d$ acquiring their extra overlap orientations in $K(d, d')$'s census rather than double-counted across kernels. The triple term dominates only where the pinned sum satisfies $\sum(1 - |h|/H)K_4 = o(\log x)$; the dominance range is stated as an explicit function class: for $H = \lambda(\log x)^\alpha$, any fixed $\lambda > 0$ and any $\alpha \geq 1$ with $\alpha > A$ —the inequality on the exponents that makes $\max(d, d') = o(H)$ hold uniformly for all $d, d' \leq (\log x)^A$ —the per-shift centred average grows only polylogarithmically in H (the Conjecture-2 scale), so $\sum(1 - |h|/H)K_4/H = O((\log \log x)^{2+\epsilon}) = o(\log x)$ and the triple term dominates throughout the polylogarithmic window class; the conjecture asserts the two-term form on exactly that class and takes no position outside it. When $H/\log^2 x \rightarrow \lambda$ the window counts converge jointly to Poisson laws with these vanishing covariances, and when $H/\log^2 x \rightarrow \infty$ every fixed finite collection $\{Z_{d_1}, \dots, Z_{d_r}\}$ is asymptotically jointly normal with independent limiting coordinates: both kernels are finite- x corrections to independence (test below), not nondegenerate limiting kernels. Both distributional conclusions are asserted under an explicit hypothesis beyond the displayed covariances, namely uniform Hardy–Littlewood tuple estimates on the window class fixed above, strong enough to control all higher joint cumulants of the window counts: every joint cumulant of order at least three vanishes in the respective normalizations. The diagonal correction to $\text{Var}_t(\pi_d)$ is Conjecture 2's pinned average, stated there.

Conjecture 9 (The sub-diffusivity of balanced prime races; apparently new). *Model a balanced race (Conjecture 1(ii)'s symmetric class differences) as the walk of its class-assignment steps, the process defined at event index: enumerate the pattern occurrences $n_1 < n_2 < \dots$ in the two competing classes, set $\xi_i = \pm 1$ by class membership of n_i , and let $\rho_k(x)$ denote the lag- k sample autocorrelation of (ξ_i) over the events with $n_i \leq x$. Then the walk is measurably sub-diffusive: (i) [decaying-repulsion law] the step autocorrelations are negative at small lags—the race analogue of the Lemke Oliver–Soundararajan consecutive-pattern repulsion [24], which supplies the mechanism—and, since that repulsion is scale-dependent rather than fixed, the conjectured form is a decay law*

$$\rho_k(x) = -\frac{c_k \log \log x + d_k}{\log x} (1 + o(1)),$$

with constants c_k, d_k in principle derivable from the LOS correlation series (underived here; and the measured values $\rho_1 \approx -0.037$, $\rho_2 \approx -0.009$ at 10^9 —universal across twin and cousin races mod 5 and 8 to within 0.004, each ~ 50 standard errors from zero—are observations at one height, consistent with $c_1 \approx 0.25$ there, not asymptotic constants); (ii) [long memory, measured] short lags do not exhaust the deficit: the running maxima $M(x) = \max_{t \leq x} |D(t)|/\sqrt{\text{count}}$ of the four races

land at quantiles 0.02–0.17 of the simulated independent-step null (median 2.34 at this span), and still only 0.03–0.23 after correcting for $\rho_{1,2}$: the negative correlation persists across lags, so the cumulative diffusivity $\sigma^2(x) = 1 + 2 \sum_k \rho_k(x)$ sits well below 1 at this height. The model layer is declared: (ξ_i) is a deterministic, nonstationary family and the $\rho_k(x)$ are empirical statistics of it, so the Green–Kubo identity $\sigma^2 = 1 + 2 \sum_k \rho_k$ is invoked under an explicit local-stationarity model hypothesis—the step process restricted to a window $[x, 2x]$ is modelled as a stationary sequence whose autocorrelations are the measured $\rho_k(x)$ —and the variance statement can equivalently be taken as a direct definition of $\sigma^2(x)$ through block averages of $(S_{n+m} - S_n)^2/m$ over the events up to x , a form in which no stationarity is assumed at all. The Green–Kubo sum requires more than the fixed-lag law of (i): the per-lag decay controls each $\rho_k(x)$ but not the sum unless the decay is uniform in k with summable tail, since the number of relevant lags may grow with x . We therefore state the needed hypothesis as part of the clause: $\rho_k(x)$ obeys the law of (i) uniformly for $k \leq K(x)$ with $\sum_{k > K(x)} |\rho_k(x)| = o(\log \log x / \log x)$ for some $K(x) \rightarrow \infty$, and—since the tail bound alone leaves the growing head sum uncontrolled—the repulsion constants are summable, $\sum_k c_k < \infty$ and $\sum_k |d_k| < \infty$, so that $\sigma^2(x) = 1 - 2(\sum_k c_k \log \log x + \sum_k d_k) / \log x (1 + o(1))$ is a genuine consequence rather than an unjustified interchange of limits. The measured correlation length at 10^9 (a few tens of lags, with $|\rho_k|$ decreasing geometrically in the observable range) is consistent with both hypotheses. Under them, $\sigma^2(x)$ returns to 1 asymptotically, making sub-diffusivity a finite-height phenomenon of $\log \log x / \log x$ size, like every other second-order law in this paper; (iii) [registered dichotomy, open] asymptotically the two possibilities are mutually exclusive, and which of them holds is the open question. Either the fixed-lag expansion of (i) extends uniformly in k with summable constants—in which case the hypotheses of (ii) hold, (ii) applies, and the race is asymptotically diffusive with $\sigma^2(x) \rightarrow 1$ (running maxima $\sim \sqrt{2 \log \log x}$, Darling–Erdős class)—or that expansion fails at long lags, and only in that case can the race inherit the almost-periodic rigidity that Rubinstein–Sarnak-type structure [6] forces on classical races (maxima $\asymp (\log \log \log x)^A$, the Montgomery–Ng class). Rigidity therefore requires a failure of the hypotheses of (ii) at long lags, and cannot coexist with them. No explicit formula is known for twin patterns, the two branches differ only beyond any computable height, and we register the dichotomy without choosing—what is conjectured is the finite-height law (i)–(ii).

3 Family laws and second-order refinements

Conjecture 10 (Uniform quadratic de Polignac; family law apparently unstated, the $d = 2$ instance previously stated, cf. OEIS A080149, [32]). For even $d \geq 2$ let $\pi_d^q(N) = \#\{n \leq N : n^2 + 1, n^2 + 1 + d \text{ both prime}\}$ and $C(d) = C(x^2 + 1, x^2 + 1 + d)$. Then: (i) [uniform family law] for every fixed $B > 0$, $\pi_d^q(N) = C(d) I_d(N) (1 + o(1))$ uniformly over even $d \leq (\log N)^B$; (ii) [registered rate, strictly stronger] there is $\eta_B > 0$ with

$$\sup_{\substack{2 \leq d \leq (\log N)^B \\ 2|d}} \left| \frac{\pi_d^q(N)}{C(d) I_d(N)} - 1 \right| \ll_B (\log N)^{-\eta_B}$$

—an error clause that no version of Bateman–Horn supplies, registered separately from (i) since its analytic content is independent; (iii) [family statistic: the law of the constants] each local factor of $C(d)$ depends only on $d \bmod p$, and the factors at any finite set of primes are exactly independent as d varies (CRT), with the passage to the infinite product controlled by the convergent tail-variance sum—which supplies the uniform integrability (via L^2 -boundedness of the normalized

partial products) that the moment identities require, since almost-sure convergence alone would not carry means through the limit; so the moments of $C(d)$ over even $d \leq D$ converge, as $D \rightarrow \infty$, to derived Euler products; in particular

$$\frac{1}{\#} \sum_{2|d \leq D} C(d) \rightarrow \bar{C} = 2.7447\dots, \quad \text{sd}(C(d)) \rightarrow 1.6835\dots,$$

and the empirical distribution of $C(d)$ converges to the law of the random product $\prod_p f_p(U_p)$ with independent uniform residues U_p . In particular ($d = 2$, the classical instance): $\#\{n \leq N : n^2 + 1, n^2 + 3 \text{ both prime}\} \sim C(2) I(N)$, $C(2) = 2.954014\dots$

Conjecture 11 (First occurrences of prime gaps, liminf form; the family is classical, the quantified form apparently unstated). *Let $\mathcal{G}$ be the set of even g that occur as gaps between consecutive primes (all even g , under Polignac's conjecture), and for $g \in \mathcal{G}$ let $p(g)$ be the prime starting the first such gap. Then: (i) $\log p(g)/\sqrt{g}$ is bounded between positive constants on $\mathcal{G}$; (ii) [liminf law, stated as the dual of the maximal-gap limsup, conditional on the realization hypothesis below]*

$$\liminf_{\substack{g \rightarrow \infty \\ g \in \mathcal{G}}} \frac{\log p(g)}{\sqrt{g}} = \sqrt{e^\gamma/2} = 0.943682\dots$$

The inversion of Granville's $\limsup G(X)/\log^2 X = 2e^{-\gamma}$ into first-occurrence coordinates is not automatic: the limsup controls how large maximal gaps get, not which gap sizes are realized as first occurrences near the envelope. Clause (ii) is therefore conditional on the realization hypothesis: infinitely many $g \in \mathcal{G}$ first occur inside a maximal-gap event at the Cramér–Granville envelope scale, i.e. with $g = (2e^{-\gamma} + o(1)) \log^2 p(g)$. Under that hypothesis the displayed value is forced by algebra; without it only the lower bound $\liminf \log p(g)/\sqrt{g} \geq \sqrt{e^\gamma/2}$ follows from the envelope. The value is registered as that dual (against the slope-1 laws), not claimed as an independent phenomenon; (iii) [second-order, singular-series dependence] sampling g uniformly from the realized gaps $\mathcal{G} \cap [G, 2G]$ and letting $G \rightarrow \infty$,

$$\log p(g) = \sqrt{g} + \frac{1}{2} \log g - \frac{1}{2} \log \mathfrak{S}^*(g) + E_g, \quad \mathfrak{S}^*(g) = \prod_{\substack{p|g \\ p > 2}} \frac{p-1}{p-2},$$

where the error E_g converges in distribution: for the sampled g ,

$$\Pr[E_g \leq t] \rightarrow 1 - \exp(-e^{2(t-\mu)}) \quad (G \rightarrow \infty)$$

for some centring constant μ —the min-type (reversed) Gumbel law at scale $\frac{1}{2}$, i.e. the law of $\frac{1}{2} \log W$ for W a unit exponential. The scale is forced by the hazard computation: the cumulative intensity of gap- g events is $\Lambda_g(y) \asymp (\mathfrak{S}(g)/y^2) \exp(y - g/y)$ in $y = \log x$ (the factor $e^{-g/y}$ is the no-intervening-prime probability), so $\frac{d}{dy} \log \Lambda_g = 1 + g/y^2 - 2/y = 2 + o(1)$ at the centring point $y_0 \approx \sqrt{g}$, and the first-passage identity $\Lambda_g(y) = W$ gives $E_g = y - y_0 = \frac{1}{2} \log W + o(1)$. The scale is $\frac{1}{2}$ and not 1 precisely because the slope of the log-intensity at y_0 is 2 (both the y -term and the g/y -term contribute), and the same slope-2 expansion is what produces the $\frac{1}{2}$ coefficients of the centring. (Gumbel-type modelling of gap extremes and first occurrences is Kourbatov–Wolf territory [15, 25]; the claim of this clause is only the $\mathfrak{S}^*$ -centring and the scale- $\frac{1}{2}$ first-passage form, not the extreme-value framework. Sampling uniformly over the realized gaps in $[G, 2G]$ conditions on realization, which the independent first-passage model does not represent; the clause

conjectures that this selection does not shift the limiting law, and that assumption is part of what a refutation would refute.) Smooth gaps, whose tuple constant is large, appear earlier by exactly half a logarithm of that constant—the waiting-time first-passage computation makes the coefficient $-\frac{1}{2}$, not a free parameter, while μ is left as the one unresolved constant of the clause. (The restriction to $\mathcal{G}$ is needed: $p(g)$ is not known to be defined for every even g , and quantifying over all g would smuggle Polignac’s conjecture into the statement. The $\mathfrak{S}^*$ -clause records that the Hardy–Littlewood factor of g materially shifts waiting times; it does not move the liminf constant, since $\log \mathfrak{S}^*(g) = O(\log \log g)$.)

Conjecture 12 (The Stern lane race; apparently new). *In Stern’s representation problem $n = p + 2k^2$ (n odd), race the k -even and k -odd lanes. Prime-square contamination of the Λ -weighted lane counts comes from $n = q^2 + 2k^2$ —the norm form $x^2 + 2y^2$ of $\mathbb{Q}(\sqrt{-2})$ with prime x —and since $q^2 \equiv 1 \pmod{8}$ while $2k^2 \equiv 0$ or $2 \pmod{8}$ by the parity of k : (i) [class assignment, direct congruence] contamination sits in the k -even lane iff $n \equiv 1 \pmod{8}$, in the k -odd lane iff $n \equiv 3 \pmod{8}$, and for $n \equiv 5, 7 \pmod{8}$ no square contamination exists at all: $q^2 + 2k^2 \equiv 1$ or $3 \pmod{8}$ always, a two-line congruence (genus theory enters only for the converse question of which n are represented), so the null classes are provable, not conjectured; (ii) [drift law, in the ensemble form of Conjecture 4] Fix the ensemble: for a scale X , let $\mathbb{E}_X$ average over odd $n \in (X, 2X]$ of the given residue class mod 8 with weight proportional to $1/n$ (the logarithmic sampling measure, as at Conjectures 5 and 4). Define, per class, the clean-minus-contaminated difference with explicit sign: $D(n) = R_o - R_e$ for $n \equiv 1 \pmod{8}$ (even lane contaminated), $D(n) = R_e - R_o$ for $n \equiv 3 \pmod{8}$, and $D(n) = R_e - R_o$ (sign immaterial) on the null classes $n \equiv 5, 7 \pmod{8}$. Set $D_{\text{sys}}(n) = [\sum_{(q,k): q^2+2k^2=n, q \text{ prime}, k \geq 1} \log q] / \lambda(n)$ —the weight is $\Lambda(q^2) = \log q$, each representation counted once in the $k \geq 1$ convention (the ordered-representation factor 2 of Conjecture 4 has no counterpart here, and doubling would be a normalization error)—where $\lambda(n)$ is the analytic lane mean of $\log p$ over the representation profile ($\lambda(n) = \int \log(n - 2u^2) d\mu_n$ under the lane measure; the verification uses the empirical lane mean as its estimator). The law, coefficient-identifying as at Conjecture 4(i): on the contaminated classes*

$$\mathbb{E}_X[D - D_{\text{sys}}] = o(\mathbb{E}_X[D_{\text{sys}}]) \quad (X \rightarrow \infty),$$

and on the null classes—where D_{sys} vanishes identically, so that a comparison against it would be empty—the statement is made against the positive comparison scale

$$A(X) := \mathbb{E}_X^{(1)}[D_{\text{sys}}],$$

the logarithmic-ensemble mean of the systematic census taken over the contaminated class $n \equiv 1 \pmod{8}$: writing D_{null} for the difference D on a null class $n \equiv 5$ or $7 \pmod{8}$, the claim is $\mathbb{E}_X[D_{\text{null}}] = o(A(X))$.

Conjecture 13 (Microscopic variance law; classical family, the constant is Montgomery—Soundararajan’s [19]). *Fix $\lambda > 0$ and $c \in (0, 1]$, and let X count primes in $[t, t + \lambda \log x)$ for t uniform on $[x, (1 + c)x]$. Then, under the Hardy–Littlewood k -tuple conjectures, $X \Rightarrow \text{Poisson}(\lambda)$ (Gallagher’s argument [18], which is conditional on exactly those conjectures), and—conditional on the quantitative hypothesis spelled out after the statement (uniform Hardy–Littlewood with $o(h)$ control of the averaged singular series), without which the display below is a registered extrapolation rather than a consequence—at finite x*

$$\frac{\text{Var } X}{\mathbb{E} X} = 1 - \frac{\log(\lambda \log x) + \gamma + \log 2\pi - 1}{\log x} + o\left(\frac{1}{\log x}\right),$$

with $\gamma + \log 2\pi - 1 = 1.41509\dots$, the leading correction being independent of the sampling constant c . [Interpolation clause] More generally, for windows of length $H = \lambda(\log x)^\alpha$ the same formula with $\log H$ in place of $\log(\lambda \log x)$,

$$\frac{\text{Var } X}{\mathbb{E} X} = 1 - \frac{\log H + \gamma + \log 2\pi - 1}{\log x} + o\left(\frac{1}{\log x}\right),$$

holds uniformly for $1 \leq \alpha \leq A$ (any fixed A with $H \leq x^{o(1)}$): one expression interpolates from Gallagher's Poisson boundary ($\alpha = 1$, deficit $\asymp \log \log x / \log x$) through the mesoscopic Montgomery–Soundararajan regime (deficit $\asymp \alpha \log \log x / \log x$), with no transition other than the slow growth of $\log H$.

Conjecture 14 (The null-mechanism race; apparently new). *For the quadratic twin pairs of Conjecture 10 at $d = 2$, square contamination is algebraically impossible: $n^2 + 1 = q^2$ has no solution with $n \geq 1$, and $n^2 + 3 = q^2$ only at $(n, q) = (1, 2)$. The surviving classes of $n \bmod 5$ are $\{0, 1, 4\}$, and the classes 1 and 4 have identical singular series (elementary, via CRT). Let $\pi_a^q(x)$ count solutions with $n \equiv a \pmod{5}$, $D(x) = \pi_1^q(x) - \pi_4^q(x)$, and define the normalized race process $Y(x) = D(x) / \sqrt{\pi_1^q(x) + \pi_4^q(x)}$. Then, in contrast to the twin race of Conjecture 1, this race is driftless: (i) $\mathcal{M}_x(D(t) \log^2 t / \sqrt{t}) \rightarrow 0$ at the drift-scale normalization of Conjecture 1(i)—the scale at which the contaminated races converge to nonzero constants—so this is the genuine negation of Conjecture 1(i) for this pair (the weaker normalization $\mathcal{M}_x(Y) \rightarrow 0$ with $Y = D / \sqrt{\pi^q}$ would not exclude a drift at the contamination scale, hence would not make this race a control; the form stated excludes exactly that); (ii) [occupation law, stated as a conditional consequence] the arithmetic content of this clause is a strong approximation hypothesis, for the process defined precisely at event index: let $n_1 < n_2 < \dots$ enumerate the n with $n^2 + 1$ and $n^2 + 3$ both prime and $n \equiv 1$ or $4 \pmod{5}$, set $\xi_i = +1$ if $n_i \equiv 1$ and -1 if $n_i \equiv 4$, and let $S_N = \xi_1 + \dots + \xi_N$ (so $D(x) = S_{N(x)}$ with $N(x)$ the number of such events to x). The sequence (ξ_i) is deterministic, so the hypothesis is stated for a randomly sited window rather than for S_N itself. Fix a growth scale N_0 and a window length $N = N(N_0)$ with $N = o(N_0)$, and choose the starting index M of the window according to the logarithmic measure on $[1, N_0]$:*

$$\Pr[M = m] = \left(\sum_{1 \leq m' \leq N_0} \frac{1}{m'} \right)^{-1} \frac{1}{m}, \quad 1 \leq m \leq N_0.$$

The hypothesis is an almost-sure invariance principle (ASIP) for the induced random path $(S_{M+j} - S_M)_{0 \leq j \leq N}$ under that sampling law: on a probability space carrying M and a standard Brownian motion B ,

$$S_{M+j} - S_M = B(j) + O(N^{1/2-\delta}) \quad \text{uniformly for } 0 \leq j \leq N,$$

almost surely in M as $N_0 \rightarrow \infty$, for some $\delta > 0$. The logarithmic sampling law is the same clock in which the occupation statement below is read, and randomizing the window start is what supplies the deterministic step sequence with a probability space. Weak convergence alone would not suffice, since a distributional limit on each fixed interval does not control pathwise occupation across unboundedly many logarithmic scales; under the ASIP the almost-sure error is absorbed at every logarithmic scale and the Lamperti/ergodic occupation argument below is valid. The weak functional limit theorem— $S_{\lfloor N\tau \rfloor} / \sqrt{N}$ converging weakly to standard Brownian motion on $\tau \in [0, 1]$ —is retained only as the weaker, separately falsifiable consequence of the ASIP.

Everything after that is probability, not number theory: granting the randomized ASIP just stated, $D(t) = B(N(t)) + O(N(t)^{1/2-\delta})$ with $N(t) \asymp t/\log^2 t$, the error being $o(\sqrt{N(t)})$ almost surely, and in logarithmic time $u = \log t$ the Lamperti reduction $Z(s) = e^{-s/2}B(e^s)$ is a stationary, ergodic Ornstein–Uhlenbeck process, so the logarithmic occupation measure of leadership converges almost surely to $\frac{1}{2}$:

$$\frac{1}{\log x} \int_2^x \mathbf{1}_{\{D(t)>0\}} \frac{dt}{t} \longrightarrow \frac{1}{2},$$

with fluctuations of the Gaussian scale $\sqrt{\log 2/\log x}$ (the occupation-indicator covariance $\frac{1}{2\pi} \arcsin e^{-u/2}$ integrates to $\frac{1}{2} \log 2$; note that the constant for the sign average $2\mathbf{1} - 1$ is twice this, the occupation fraction itself carrying $\sqrt{\log 2/\log x}$); the classical arcsine spread survives only in the natural event clock, i.e. for occupation fractions of $\{B(m) > 0, m \leq N(x)\}$. A persistent systematic leader—a log-occupation stuck near 0 or 1 beyond the stated Gaussian scale—is the refutation.

Conjecture 15 (The cubic shift family: the distribution of its constants; instance-level law apparently unstated). For non-cube $a \geq 1$ let $C(a) = C(x^3 + a)$. Only $p \equiv 1 \pmod{3}$ moves the product, and the local root count there takes three values as a varies mod p : $\omega = 3$ with probability $(p-1)/3p$ ($-a$ a nonzero cube), $\omega = 1$ with probability $1/p$ (the case $p \mid a$, easy to overlook and essential to the mean-value identity below), and $\omega = 0$ with probability $2(p-1)/3p$; for $p \equiv 2 \pmod{3}$ the cube map is a bijection and $\omega = 1$ identically. Then: (i) [mean-value one, elementary lemma] $\mathbb{E}_a[\omega(p)] = 3 \cdot \frac{p-1}{3p} + \frac{1}{p} = 1$ exactly at every p , so the derived mean of $C(a)$ is exactly 1—the cubic analogue of the Korevaar–te Riele mean-value-one theorem for linear prime-pair constants (arXiv:0806.1667), here a one-line computation at each finite level; passing the mean through the infinite product needs more than almost-sure convergence, and the needed ingredient is on hand: the convergent variance sum makes the partial products an L^2 -bounded martingale (each factor has mean 1 and is independent of the earlier ones), so they are uniformly integrable and the limit law has mean exactly 1, not merely limit-of-means 1; (ii) [limit law] as $a \rightarrow \infty$ over non-cubes, $C(a)$ converges in distribution to the random Euler product $\prod_{p \equiv 1(3)} f_p(\xi_p)$ with the three-state local variables ξ_p above—independent at any finite set of primes exactly, by CRT, with the infinite product’s tail controlled by the convergent variance sum (unlike the quadratic family of Conjecture 10(iii), no normalization is needed)—with derived standard deviation 0.2762... computed from the full three-state law; (iii) [uniformity] $\#\{n \leq N : n^3 + a \text{ prime}\} = C(a)I_a(N)(1 + o(1))$ uniformly over non-cube $a \leq (\log N)^B$.

Conjecture 16 (Conjecture F as a family; the core is previously stated, [1], cf. [23]). For odd A let $Q_A(N) = \#\{n \leq N : n^2 + n + A \text{ prime}\}$ and $C(A) = C(x^2 + x + A)$. Then, for every fixed $B > 0$, $Q_A(N) = C(A)I_A(N)(1 + o(1))$ uniformly over odd $1 \leq A \leq (\log N)^B$, with the residual field obeying the analogue of Conjecture 8(ii), kernel derived and probability space specified: for t uniform on $[N, 2N]$ and window length $H \leq N^{o(1)}$, let $Q_A(t; H) = \#\{t < n \leq t + H : n^2 + n + A \text{ prime}\}$ (mean $\asymp H/2 \log N$, which sets the regime). Two members share the index window, so with $C(A, A') = C(x^2 + x + A, x^2 + x + A')$,

$$\text{Cov}_t(Q_A, Q_{A'}) \sim [C(A, A') - C(A)C(A')] \frac{H}{4 \log^2 N},$$

with $C(A, A') = 0$ (hence negative correlation) for the locally exclusive pairs. This is the same-index contribution only: the full covariance adds the off-index pinned sum

$$\frac{1}{4 \log^2 N} \sum_{0 < |h| \leq H} (H - |h|) [C(f_A(x), f_{A'}(x + h)) - C(A)C(A')],$$

the two-parameter analogue of Conjecture 2's $G(H)$, whose evaluation is this family's open component. (The order of this sum matters: the joint event here is two single prime values at distinct indices, of probability $\asymp 1/4 \log^2 N$, unlike Conjecture 8's pair-of-pairs events at $1/\log^4 x$. At that order the off-index sum, which grows like a pinned G -average over $\asymp H$ shifts against the same-index term's single H , is generically the larger component, so the numerical cancellation of the same-index kernel establishes nothing about the full cross-member covariance, and the localization of the observed profile deficit to the diagonal is an open question rather than an inference.) For $H/\log N \rightarrow \lambda$ the counts are jointly Poisson with local exclusions, and for $H/\log N \rightarrow \infty$ each fixed finite collection of studentized counts is asymptotically jointly normal, all correlations of size $\asymp 1/\log N$ making both kernels finite- N corrections to independence, exactly as at Conjecture 8(ii). (Uniformity over a range growing with N is the substantive claim; uniformity over any fixed finite set of A is logically automatic from the individual asymptotics.) Among odd $A \leq 199$, Euler's $A = 41$ has the largest constant, $C(41) = 6.6395463 \dots$ (Cohen's high-precision value [31]).

Conjecture 17 (Twin-member Goldbach, orientation-resolved; the basis conjecture and the integral kernel are Dubner's [9]; the orientation decomposition apparently new). *For even n let $R_T(n)$ count ordered (a, b) , $a + b = n$, with a and b both members of twin pairs, each ordered role (lower/upper) counted. Each of the four role assignments is a 4-form linear system in $t = a$ with root set $\{0, \pm 2\} \cup \{n, n \pm 2\}$ -type mod each p , hence its own n -dependent singular series $\mathfrak{S}_4^{(o)}(n)$, and*

$$R_T(n) = \left[\sum_o \mathfrak{S}_4^{(o)}(n) \right] \int_5^{n-5} \frac{dt}{\log^2 t \log^2(n-t)} (1 + o(1)),$$

the four constants computed exactly per n from the coincidence pattern of the root sets (which of $n, n \pm 2, n \pm 4$ each small prime divides). The error term is stated as $(1 + o(1))$, not $(1 + O(1/\log n))$: our own measurements show a second-order deficit whose $1/\log n$ coefficient is not yet pinned down (below). Two precision notes: the asymptotic runs over varying n , so what is assumed is a Hardy–Littlewood estimate for 4-form systems uniform in the n -dependent coefficients—strictly stronger than the fixed-system conjecture—and the number 5, the unique prime that is simultaneously a lower twin member (of $(5, 7)$) and an upper one (of $(3, 5)$), is double-counted consistently by the ordered-role convention, affecting finitely many representations per n and no asymptotic.

4 Instances and structural companions

Conjecture 18 (Contamination in prime triplets; apparently new). *The contamination calculus of Conjecture 1(v) extends to k -tuples, and the triplet pattern $(n, n + 2, n + 6)$ is its first instance beyond pairs. Of the three configurations placing a prime square inside the pattern, two die algebraically for $q > 3$ — $(q^2, q^2 + 2, q^2 + 6)$ by $3 \mid q^2 + 2$, and $(q^2 - 6, q^2 - 4, q^2)$ because $q^2 - 4 = (q - 2)(q + 2)$ is composite (at $q = 3$ the factorization is trivial, $q^2 - 4 = 5$ is prime, and the single bounded exceptional configuration $(3, 5, 9)$ occurs—one term, harmless to every asymptotic)—leaving the single doubly-thinned survivor $(q^2 - 2, q^2, q^2 + 4)$, which requires $q^2 - 2$ and $q^2 + 4$ simultaneously prime and forces $q \equiv \pm 2 \pmod{5}$ for $q \neq 5$ (else $5 \mid q^2 + 4$; at $q = 5$ all three conditions do hold, the configuration being $(23, 25, 29)$, but its start $23 \equiv 3 \pmod{5}$ lies outside the triplet start classes $\{1, 2\}$, so the single term is harmless). Triplet starts lie in classes $n \equiv 1, 2 \pmod{5}$, and the surviving configuration has start $q^2 - 2 \equiv 2$: class 2 is contaminated*

and class 1 leads, in the drift-scale form of Conjecture 1(i) (a normalization at the noise scale, $\mathcal{M}_x((D - T_3)/\sqrt{\pi_3})$, would be vacuous for the coefficient, exactly as at the pair races):

$$\mathcal{M}_x\left(\frac{(\pi_3(t; 5, 1) - \pi_3(t; 5, 2)) \log^3 t}{\sqrt{t}}\right) \rightarrow c_3 := \lim_{x \rightarrow \infty} \frac{T_3(x) \log^3 x}{\sqrt{x}}, \quad T_3(x) = \frac{1}{\log^3 x} \sum_{\substack{q \leq \sqrt{x}, q \neq 5 \\ q^2-2, q^2+4 \text{ prime}}} \log(q^2-2) \log q \log$$

(c_3 exists and is the Bateman–Horn constant of the triple $(q, q^2 - 2, q^2 + 4)$, since the census sum is $\sim c_3 \sqrt{x}$; convergence of the logarithmic mean at this normalization is the same Rubinstein–Sarnak-type structure hypothesis as at Conjecture 1(i)), with drift-to-noise $\asymp \log^{-3/2} x$ —one logarithm weaker than the pair races, a scale the calculus itself predicts and the verification must respect. The mechanism clause and the averaging clause are asserted separately, as at Conjecture 1(i).

Conjecture 19 (The contamination matrix: sexy pairs with two surviving orientations; apparently new). For the pattern $(n, n + 6)$, both prime-square orientations survive, on complementary classes of $q \bmod 5$ —the first matrix instance of the calculus (every pair race so far had exactly one): orientation $A = (q^2 - 6, q^2)$ requires $q^2 - 6$ prime, forcing $q \equiv \pm 2 \pmod{5}$ for $q \neq 5$ (at $q = 5$ the value 19 is prime, but its start $19 \equiv 4 \pmod{5}$ lies outside the sexy start classes $\{1, 2, 3\}$), and lands in start class 3 (mod 5), class 3 (mod 8); orientation $B = (q^2, q^2 + 6)$ requires $q^2 + 6$ prime, forcing $q \equiv \pm 1 \pmod{5}$ (the term $q = 5$ falls outside the start classes), and lands in class 1 (mod 5), class 1 (mod 8). Sexy starts occupy classes $\{1, 2, 3\} \bmod 5$ and $\{1, 3, 5, 7\} \bmod 8$; with $T_A(x) = \frac{1}{\log^2 x} \sum_{q^2-6 \text{ prime}, q \neq 5} \log q \log(q^2 - 6)$, $T_B(x) = \frac{1}{\log^2 x} \sum_{q^2+6 \text{ prime}, q \neq 5} \log q \log(q^2 + 6)$, and c_A, c_B their drift-scale limits ($c_A = \lim T_A \log^2 x / \sqrt{x}$, the Bateman–Horn constant of $(q, q^2 - 6)$, likewise c_B for $(q, q^2 + 6)$), the predicted drift vector, in the drift-scale limiting-log-mean sense of Conjecture 1(i) (an $\mathcal{M}_x$ -over- $\sqrt{\pi}$ normalization would be coefficient-blind), is: mod 5: $\mathcal{M}_x((\pi(2) - \pi(3)) \log^2 t / \sqrt{t}) \rightarrow c_A$, $\mathcal{M}_x((\pi(2) - \pi(1)) \log^2 t / \sqrt{t}) \rightarrow c_B$ (class 2 clean); mod 8: $\mathcal{M}_x((\frac{1}{2}(\pi(5) + \pi(7)) - \pi(3)) \log^2 t / \sqrt{t}) \rightarrow c_A$, $\mathcal{M}_x((\frac{1}{2}(\pi(5) + \pi(7)) - \pi(1)) \log^2 t / \sqrt{t}) \rightarrow c_B$ (classes 5, 7 clean, mutually symmetric, with $\mathcal{M}_x((\pi(5) - \pi(7)) \log^2 t / \sqrt{t}) \rightarrow 0$). Two independently computable drift constants, one certified null class per modulus. The mechanism clause and the averaging clause are asserted separately, as at Conjecture 1(i).

Conjecture 20 (Fibonacci–Lucas twins: the convergent side stress-tested; the object is OEIS A080327). (i) [rank-disjointness, elementary, with its exception stated] every odd prime factor r of L_p has rank of apparition exactly $2p$ (from $z(r) \mid 2p$, $z(r) \nmid p$, and $z(r) = 2$ being impossible), and every prime factor of F_p ($p \neq 5$ prime) has rank exactly p : the odd divisor pools of the two numbers are disjoint. The prime 2 is the unique exception: 2 divides both F_p and L_p precisely when $p = 3$ ($F_3 = 2$, $L_3 = 4$; $\gcd(F_p, L_p) = 2$ iff $3 \mid p$)—the restriction to odd divisors is essential, the clause being false at $p = 3$ without it. Disjoint pools remove shared-prime correlation but not all correlation: dependence can still enter through the order structure of the recurrence, which is why (iii) matters; (ii) [finiteness] only finitely many primes p have F_p and L_p simultaneously prime; (iii) [calibration clause, quantitative] the naive joint accounting—probability $(e^\gamma \log p / (p \log \phi))^2$ per index—assigns the catalogued index $p = 148091$ (OEIS A080327; both F_p and L_p are probable primes—numbers of roughly 30,950 digits, with Lucas numbers proved prime only up to index 56003—so everything recorded here is conditional on both probable-prime classifications, for which no Baillie–PSW pseudoprime is known: no unconditional refutation is claimed from uncertified numbers) prior mass about 3×10^{-3} beyond 10^4 . That rare-event datum is recorded descriptively, with no significance claim attached to it: the model was examined after the event was known,

and no test protocol was fixed in advance. A single surprising event is not a strict refutation, and it does not by itself estimate the size of the correction: an underestimated constant, genuine cross-dependence, and a mis-calibrated finite-range tail are competing explanations that one event cannot separate. What the datum does bear on (conditionally as above) is not the model but the list: it is fatal to any Goldilocks completeness list; the conjectural content of (ii) is that even the corrected accounting has convergent sum.

Conjecture 21 (Factorial twins; the uniqueness core is previously stated, OEIS A088054; clauses (i) and (iii) are ours). (i) [window rigidity, elementary] for $n \geq 4$ and $2 \leq |a| \leq n$ the number $n! + a$ is composite (any prime factor p of a satisfies $p \leq n$, so $p \mid n! + a$, and $n! + a > n \geq p$); the two small anomalies $2! - 2 = 0$ and $3! - 3 = 3$ are the only exceptions, which is why the clause is restricted to $n \geq 4$. Hence among all offsets of bounded size the only candidates for primality near $n!$ are $a = \pm 1$: the twin question below is the unique bounded-offset constellation question at $n!$, and every admissible offset set is a subset of $\{-1, +1\}$. (ii) [uniqueness; previously stated] Only finitely many n have $n! - 1$ and $n! + 1$ simultaneously prime, and the complete list is $n = 3$: i.e. $6 = 3! = 3\#$ is the unique factorial lying between twin primes. (iii) [joint fluctuation model] the two single-sided counts $F_{\pm}(N) = \#\{n \leq N : n! \pm 1 \text{ prime}\}$ are each $\sim e^{\gamma} \log N$ and, under the independent-indices model stated below, asymptotically independent, with $F_+ - F_-$ normalized by $\sqrt{2e^{\gamma} \log N}$ asymptotically standard normal. The probability space is declared: the clause is about a random model in which the events $E_n^{\pm} = \{n! \pm 1 \text{ prime}\}$ carry the Caldwell–Gallot marginal hazards e^{γ}/n with an unspecified dependence structure, and the Gaussian limit is asserted under three explicit hypotheses on that structure: the same-index bound $\sum_{n \leq N} |\text{Cov}(\mathbf{1}_{E_n^+}, \mathbf{1}_{E_n^-})| = o(\log N)$, which pins the variance at $2e^{\gamma} \log N (1 + o(1))$ and which the joint law $\sim (e^{\gamma}/n)^2$ derived below supplies with a convergent sum—without it the remaining hypotheses leave the second cumulant free, and a model with $\Pr(E_n^+ \cap E_n^-) = e^{\gamma}/2n$ halves the limiting variance; the cross-index covariance bound $\sum_{m < n \leq N} |\text{Cov}(\mathbf{1}_{E_m}, \mathbf{1}_{E_n})| = o(\log N)$ over the four sign pairs (the variance itself is $\asymp \log N$), and—since pairwise control alone cannot preclude higher-order dependence—the higher-cumulant condition $\kappa_r(F_+ - F_-) = o((\log N)^{r/2})$ for each fixed $r \geq 3$, the method-of-moments sufficient condition. Neither is imported from an independence model; whether they hold for the arithmetic dependence across factorial indices is the open component of this clause.

Theorem 1 (Cube obstruction). For $k \geq 2$, the cube k^3 is representable as $p + j^3$ with p prime, $j \geq 1$, if and only if $3k^2 - 3k + 1$ is prime. Consequently the set of integers not representable as $p + k^3$ is infinite: the non-representable cubes alone have counting function $\sim x^{1/3}$. (That the full exceptional set is $\sim x^{1/3}$ follows only when combined with Conjecture 22(i), and is conjectural.)

Proof. $k^3 - j^3 = (k - j)(k^2 + kj + j^2)$; for $1 \leq j \leq k - 2$ both factors exceed 1, so the difference is composite. The only candidate is $j = k - 1$, giving $3k^2 - 3k + 1$. Since $3k^2 - 3k + 1$ is composite for a density-one set of k (all but $O(K/\log K)$ of $k \leq K$, by an upper-bound sieve of Brun or Selberg type applied to the quadratic's prime values), all but a vanishing proportion of cubes are unrepresentable. $\square$

Conjecture 22 (The boundary trichotomy for polynomial ladders; the divisibility principle is classical and its cubic case is Cunningham's cuban observation (OEIS A002407); the family classification is apparently new). For $F \in \mathbb{Z}[x]$ with $\deg F \geq 2$ and positive leading coefficient, and $m > j \geq 1$, $(m - j) \mid F(m) - F(j)$ (textbook), and the divided difference $(F(m) - F(j))/(m - j)$ —whose leading form is then positive on the cone $m > j \geq 1$ —exceeds 1 outside an effectively

bounded region; so $F(m) - F(j)$ prime forces $m - j = 1$ apart from finitely many exceptional pairs (for the family below the exceptional set is provably empty). Both hypotheses are necessary: for $\deg F = 1$ the cofactor can be identically 1 and the collapse fails, and for negative leading coefficient the divided difference tends to $-\infty$ on the cone, so the “exceeds 1” clause would be false as stated (one may equivalently keep general sign and read the claim through $|F(m) - F(j)|$). Every representation problem $F(m) = p + F(j)$ in this range thus collapses to the boundary polynomial $D_F(m) = F(m) - F(m - 1)$. The content is the classification of the boundary lanes. For $F = x^3 + cx$, $c \geq 0$: (i) [trichotomy, elementary] $D_F = 3m^2 - 3m + 1 + c$ and the lane is dead-parity for c odd (D_F always even), dead-3-adic for even $c \equiv 2 \pmod{3}$ ($3 \mid D_F$ always), and admissible exactly for $c \equiv 0, 4 \pmod{6}$; (ii) [uniform boundary law] over the admissible lanes, $\#\{m \leq M : D_F(m) \text{ prime}\} \sim C(D_F) I(M)$ uniformly for $c \leq (\log M)^B$ —a Bateman–Horn family whose members are indexed by the ladder classification; (iii) [attributed core] the case $c = 0$ is the cuban ladder: Cunningham’s classical observation that a prime difference of cubes forces consecutive arguments, with the non-cube exceptional set of $n = p + k^3$ finite (Hardy–Littlewood’s $E_3(X) = O(1)$ [1]).

Conjecture 23 (The power-obstruction ladder: an elementary structural proposition with a Bateman–Horn corollary; apparently new as a family). For $k \geq 2$ and $m \geq 2$, write $D_k(m) = m^k - (m - 1)^k$, and call m^k representable if $m^k = p + j^k$ for some prime p and $j \geq 1$. Then: (i) [theorem] for composite k , no k -th power is representable: if r is a proper divisor of k with $r > 1$ (such r exists precisely because k is composite; the excluded value $r = 1$ would allow only the useless factor $m - j = 1$ at $j = m - 1$), then for every $1 \leq j < m$ the difference $m^k - j^k$ has the proper factor $m^r - j^r > 1$ with cofactor > 1 ; (ii) [elementary] for prime k (including $k = 2$), m^k is representable if and only if $D_k(m)$ is prime; and D_k is irreducible for prime k (a root α has $(\alpha/(\alpha - 1))^k = 1$ with $\alpha/(\alpha - 1) \neq 1$, so $\zeta = \alpha/(\alpha - 1)$ is a primitive k -th root of unity and $\mathbb{Q}(\alpha) = \mathbb{Q}(\zeta)$ has degree $k - 1 = \deg D_k$, forcing irreducibility; explicitly $D_k(x) = (x - 1)^{k-1} \Phi_k(x/(x - 1))$. The forcing $j = m - 1$ needs only $k \geq 2$); (iii) [conjecture] for each prime $k \geq 3$ the representable lane follows Bateman–Horn ($k = 2$ is a theorem, the prime number theorem on the lane $2m - 1$): $\#\{m \leq M : D_k(m) \text{ prime}\} \sim C_k \int_2^M dt / \log D_k(t)$).

Conjecture 24 (The alternating cyclotomic chain; apparently new). There are infinitely many primes p such that, writing $u = \Phi_3(p) = p^2 + p + 1$, the three integers

$$p, \quad u = p^2 + p + 1, \quad \Phi_6(u) = u^2 - u + 1 = p^4 + 2p^3 + 2p^2 + p + 1$$

are all prime, with $\#\{p \leq x\} \sim C_6^{\text{ch}} I(x)$ for the Bateman–Horn system $\{x, x^2 + x + 1, x^4 + 2x^3 + 2x^2 + x + 1\}$, $C_6^{\text{ch}} = 3.6143 \pm 0.011$. The first chains are $(2, 7, 43)$ and $(3, 13, 157)$.

Conjecture 25 (Twin cyclotomic bases: the complete quadratic family; no prior trace found). For the three cyclotomic polynomials of degree 2 (indices k with $\varphi(k) = 2$, i.e. $k \in \{3, 4, 6\}$), consider $\Phi_k(n)$ and $\Phi_k(n + 1)$ simultaneously prime. Then: (i) [parity obstruction, elementary] for $k = 4$ one member of the pair $(n^2 + 1, n^2 + 2n + 2)$ is even for every n , so the only prime pair is $n = 1$, giving $(2, 5)$ —the same obstruction that forbids consecutive $n^2 + 1$ primes; (ii) [collapse, elementary] $\Phi_6(x) = \Phi_3(x - 1)$ identically, so the $k = 6$ pair at index n is the $k = 3$ pair at index $n - 1$: the family contains exactly one nontrivial statement; (iii) [conjecture, the live instance] there are infinitely many n for which the length-3 repunits in bases n and $n + 1$ are simultaneously prime: $\#\{n \leq N : \Phi_3(n) = n^2 + n + 1 \text{ and } \Phi_3(n + 1) = n^2 + 3n + 3 \text{ both prime}\} \sim C_5 I(N)$, with $C_5 = C(x^2 + x + 1, x^2 + 3x + 3)$.

References

- [1] G. H. Hardy and J. E. Littlewood, *Some problems of ‘Partitio numerorum’; III: On the expression of a number as a sum of primes*, Acta Math. 44 (1923), 1–70. DOI: 10.1007/BF02403921.
- [2] P. T. Bateman and R. A. Horn, *A heuristic asymptotic formula concerning the distribution of prime numbers*, Math. Comp. 16 (1962), 363–367. DOI: 10.1090/S0025-5718-1962-0148632-7.
- [3] D. R. Heath-Brown, *Primes represented by $x^3 + 2y^3$* , Acta Math. 186 (2001), 1–84. DOI: 10.1007/BF02392715.
- [4] C. K. Caldwell and Y. Gallot, *On the primality of $n! \pm 1$ and $2 \times 3 \times 5 \times \cdots \times p \pm 1$* , Math. Comp. 71 (2002), 441–448. DOI: 10.1090/S0025-5718-01-01315-1.
- [5] G. Lillie, *About the primality of primorials*, arXiv:2110.04302 [math.NT] (2021).
- [6] M. Rubinstein and P. Sarnak, *Chebyshev’s bias*, Experiment. Math. 3 (1994), no. 3, 173–197. DOI: 10.1080/10586458.1994.10504289.
- [7] J. Grantham and A. Granville, *Fibonacci primes, primes of the form $2^n - k$ and beyond*, J. Number Theory 261 (2024), 190–219; arXiv:2307.07894. DOI: 10.1016/j.jnt.2024.02.002.
- [8] S. S. Wagstaff, Jr., *Divisors of Mersenne numbers*, Math. Comp. 40 (1983), 385–397. DOI: 10.1090/S0025-5718-1983-0679454-X.
- [9] H. Dubner, *Twin prime conjectures*, J. Recreational Math. 30, no. 3 (1999–2000) (article reproduced at OEIS A007534).
- [10] H. Dubner, *Carmichael numbers of the form $(6m + 1)(12m + 1)(18m + 1)$* , J. Integer Seq. 5 (2002), Article 02.2.1.
- [11] E. Kowalski, *Averages of Euler products, distribution of singular series and the ubiquity of Poisson distribution*, Acta Arith. 148 (2011), no. 2, 153–187; arXiv:0805.4682.
- [12] T. Oliveira e Silva, S. Herzog, and S. Pardi, *Empirical verification of the even Goldbach conjecture and computation of prime gaps up to $4 \cdot 10^{18}$* , Math. Comp. 83 (2014), no. 288, 2033–2060. DOI: 10.1090/S0025-5718-2013-02787-1.
- [13] K. Martin, *Refined Goldbach conjectures with primes in progressions*, Exp. Math. 31 (2022), 226–232; arXiv:1806.00946. DOI: 10.1080/10586458.2019.1596849.
- [14] D. A. Goldston and A. H. Ledoan, *Jumping champions and gaps between consecutive primes*, Int. J. Number Theory 7 (2011), 1413–1421; arXiv:0910.2960. DOI: 10.1142/S179304211100471X.
- [15] A. Kourbatov, *Maximal gaps between prime k -tuples: a statistical approach*, J. Integer Seq. 16 (2013), Article 13.5.2; arXiv:1301.2242. See also arXiv:1309.4053.
- [16] D. Shanks, *On maximal gaps between successive primes*, Math. Comp. 18 (1964), 646–651. DOI: 10.1090/S0025-5718-1964-0167472-8.
- [17] A. Granville, *Harald Cramér and the distribution of prime numbers*, Scand. Actuar. J. 1995:1, 12–28. DOI: 10.1080/03461238.1995.10413946.
- [18] P. X. Gallagher, *On the distribution of primes in short intervals*, Mathematika 23 (1976), 4–9. DOI: 10.1112/S0025579300016442.
- [19] H. L. Montgomery and K. Soundararajan, *Primes in short intervals*, Comm. Math. Phys. 252 (2004), 589–617. DOI: 10.1007/s00220-004-1222-4.

- [20] T. Haddad, S.-K. Leung, and C. Sabuncu, *Visiting early at prime times*, arXiv:2408.11781 (2024). DOI: 10.48550/arXiv.2408.11781.
- [21] S. S. Wagstaff, Jr., *Greatest of the least primes in arithmetic progressions having a given modulus*, Math. Comp. 33 (1979), 1073–1080. DOI: 10.1090/S0025-5718-1979-0528061-7.
- [22] R. Crandall, K. Dilcher, and C. Pomerance, *A search for Wieferich and Wilson primes*, Math. Comp. 66 (1997), 433–449. DOI: 10.1090/S0025-5718-97-00791-6.
- [23] M. J. Jacobson, Jr. and H. C. Williams, *New quadratic polynomials with high densities of prime values*, Math. Comp. 72 (2003), 499–519. DOI: 10.1090/S0025-5718-02-01418-7.
- [24] R. J. Lemke Oliver and K. Soundararajan, *Unexpected biases in the distribution of consecutive primes*, Proc. Natl. Acad. Sci. USA 113 (2016), E4446–E4454; arXiv:1603.03720. DOI: 10.1073/pnas.1605366113.
- [25] A. Kourbatov and M. Wolf, *On the first occurrences of gaps between primes in a residue class*, J. Integer Seq. 23 (2020), Article 20.9.3; arXiv:2002.02115.
- [26] M.-A. Sanchis-Lozano, *A heuristic study of the distribution of primes in short and not-so-short intervals*, arXiv:1804.07659 (2018). DOI: 10.48550/arXiv.1804.07659.
- [27] S. Sahoo, *On twin prime distribution and associated biases*, arXiv:2111.09053 (2021). DOI: 10.48550/arXiv.2111.09053.
- [28] W. Puskarczyk, *Statistical bias in the distribution of prime pairs and isolated primes*, arXiv:1807.00406 (2018). DOI: 10.48550/arXiv.1807.00406.
- [29] S.-K. Leung, *Moments of primes in progressions to a large modulus*, Forum Mathematicum, published online 26 August 2024; arXiv:2402.07941.
- [30] A. Fiori, *The least prime in arithmetic an progression* [sic], arXiv:2404.02329 (2024). DOI: 10.48550/arXiv.2404.02329.
- [31] H. Cohen, *High precision computation of Hardy–Littlewood constants*, unpublished notes (PDF hosted at OEIS A221712).
- [32] N. A. Carella, *Twin primes in quadratic arithmetic progressions*, arXiv:1710.07827 (unrefereed preprint). DOI: 10.48550/arXiv.1710.07827.